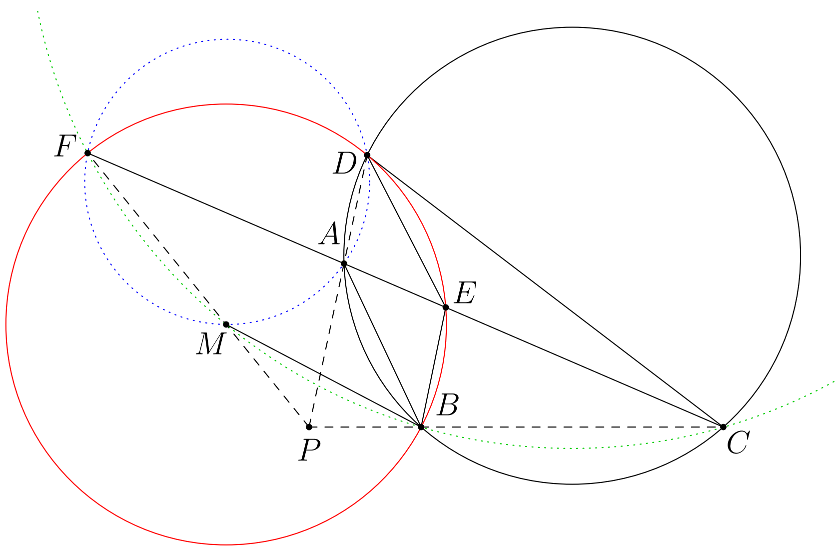
## Radical Axes — Solutions

**Problem 1.** (MEMO 2010 Individual Problem 3) We are given a cyclic quadrilateral $ABCD$ with a point $E$ on the diagonal $AC$ such that $AD = AE$ and $CB = CE$. Let $M$ be the centre of the circumcircle $k$ of the triangle $BDE$. The circle $k$ intersects the line $AC$ in the points $E$ and $F$. Prove that the lines $FM$, $AD$ and $BC$ meet at one point.

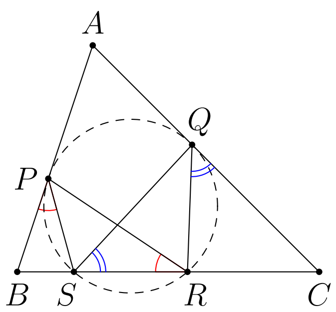



**Solution.** As $M$ is the centre of $(BEF)$, using directed angles, we obtain

$$\measuredangle BMF = 2\measuredangle BEF = 2\measuredangle BEC = \measuredangle BEC + \measuredangle CBE = \measuredangle BCE = \measuredangle BCF.$$

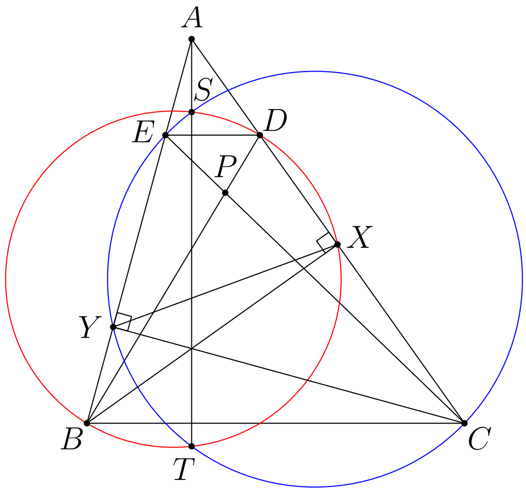
This implies $B, C, F, M$ are concyclic. By symmetry, $A, D, F, M$ are concyclic. So $FM, AD, BC$ are concurrent at the radical centre of $(BCFM)$, $(ADFM)$ and $(ABCD)$. $\square$

**Problem 2.** (USAJMO 2012 Problem 1) Given a triangle $ABC$, let $P$ and $Q$ be points on segments $AB$ and $AC$, respectively, such that $AP = AQ$. Let $S$ and $R$ be distinct points on segment $BC$ such that $S$ lies between $B$ and $R$, $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that $P, Q, R, S$ are concyclic.



**Solution.** From $\angle BPS = \angle PRS$, we see that $BP$ is tangent to $(PSR)$. From $\angle CQR = \angle QSR$, we see that $CQ$ is tangent to $(QSR)$. We are done if the two circles are the same circle. Suppose on the contrary that they are different circles. Then $SR$ is their radical axis. However, since $AP = AQ$ and $AP, AQ$ are tangent to the two circles respectively, the point $A$ lies on the radical axis $SR$. This is impossible. Therefore, $P, Q, R, S$ are concyclic. $\square$

**Problem 3.** (Thailand MO 2021 Problem 8) Let $P$ be a point inside an acute triangle $ABC$. Let the lines $BP$ and $CP$ intersect the sides $AC$ and $AB$ at $D$ and $E$, respectively. Let the circles with diameters $BD$ and $CE$ intersect at points $S$ and $T$. Prove that if the points $A$, $S$, and $T$ are collinear, then $P$ lies on a median of $\triangle ABC$.

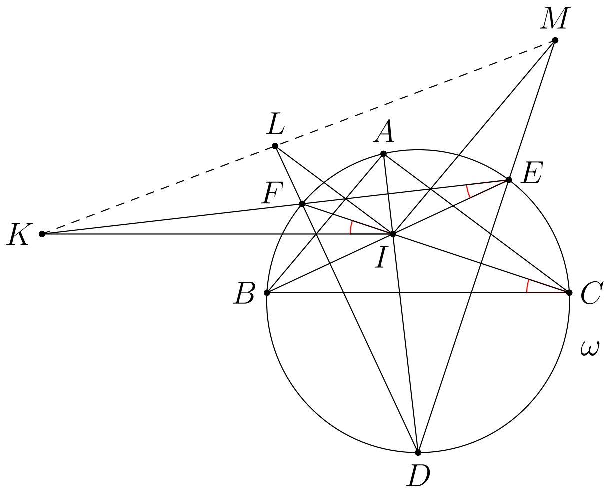



**Solution.** Let $X$ and $Y$ be the feet of altitudes from $B$ and $C$ respectively. Then $X$ lies on $(BDS)$ since $\angle BXD = 90^\circ$, and similarly $Y$ lies on $(CES)$. As $XD \cap YE = A$ lies on the radical axis $ST$ of the two circles, the points $E, Y, X, D$ are concyclic. As $B, C, X, Y$ are also concyclic, we have $BC // ED$ by Reim's theorem. It follows from Ceva's theorem that $BD \cap CE = P$ lies on the median from $A$. $\square$

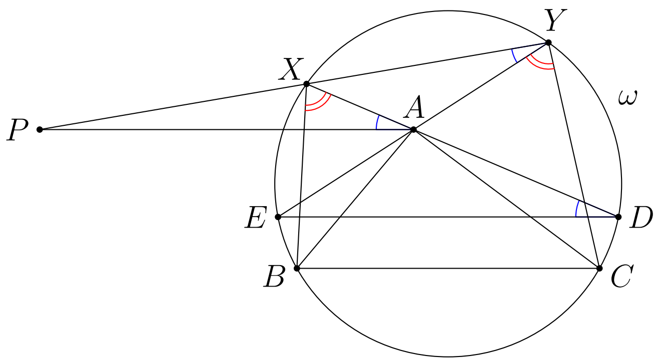
**Problem 4.** (Balkan MO 2015 Problem 2) Let $\triangle ABC$ be a scalene triangle with incentre $I$ and circumcircle $\omega$. Lines $AI, BI, CI$ intersect $\omega$ for the second time at points $D, E, F$, respectively. The parallel lines from $I$ to the sides $BC, AC, AB$ intersect $EF, DF, DE$ at points $K, L, M$, respectively. Prove that the points $K, L, M$ are collinear.

**Solution.** Note that $\angle FIK = \angle FCB = \angle FEI$. This shows $KI$ is tangent to $(EFI)$, and hence $KE \times KF = KI^2$. Therefore, the powers of $K$ with respect to $\omega$ and the point circle $I$ are equal. This implies $K$ lies on the radical axis of these circles.

Similarly, both $L$ and $M$ lie on the radical axis of these circles, so that $K, L, M$ are collinear. $\square$




**Problem 5.** (Balkan MO 2021 Problem 1) Let $ABC$ be a triangle with $AB < AC$. Let $\omega$ be a circle passing through $B, C$ and assume that $A$ is inside $\omega$. Suppose $X, Y$ lie on $\omega$ such that $\angle BXA = \angle AYC$. Suppose also that $X$ and $C$ lie on opposite sides of the line $AB$ and that $Y$ and $B$ lie on opposite sides of the line $AC$. Show that, as $X, Y$ vary on $\omega$, the line $XY$ passes through a fixed point.



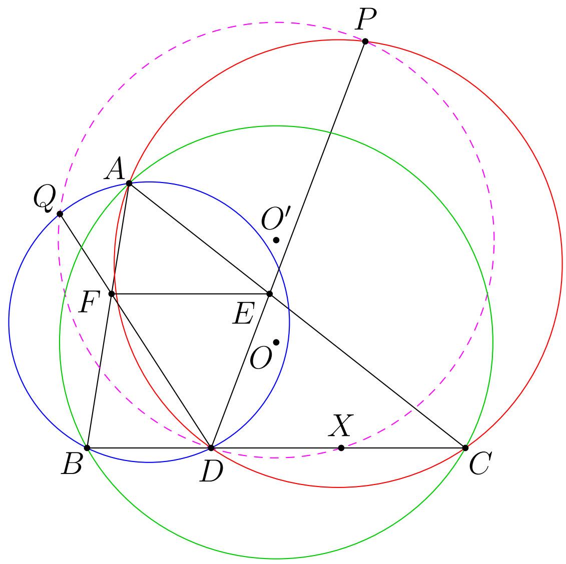

**Solution.** Let $XA$ meet $\omega$ again at $D$, and $YA$ meet $\omega$ again at $E$. From

$$\angle BXD = \angle BXA = \angle CYA = \angle CYE,$$

we know that $BC//ED$. As $AB < AC$, $XY$ and $DE$ are not parallel. Let $P$ be the point on the line $XY$ such that $PA//BC$. We claim that $P$ is a fixed point independent of $X$ and $Y$.

Indeed, note that $\angle XAP = \angle XDE = \angle XYA$. This shows $PA$ is tangent to $(AXY)$, and hence $PX \times PY = PA^2$. Therefore, the powers of $P$ with respect to $\omega$ and the point circle $A$ are equal. As $A$ cannot be the centre of $\omega$, the two circles have a radical axis $\ell$. Note that $\ell$ is different from $AP$ because $A$ is the centre of itself. Therefore, $P$ is the unique intersection point of $\ell$ and the line passing through $A$ and parallel to $BC$. Thus, $XY$ always passes through this fixed point $P$. $\square$

**Problem 6.** In $\triangle ABC$, let $D, E, F$ be points on sides $BC, CA, AB$ respectively such that $FE // BC$. The line $DE$ meets the circumcircle of $\triangle ACD$ again at $P$. The line $DF$ meets the circumcircle of $\triangle ABD$ again at $Q$. Let $X$ be the symmetric point of $D$ about the midpoint of $BC$. Prove that $P, Q, D, X$ are concyclic.




**Solution.** Let $O$ and $O'$ be the circumcentres of $\triangle ABC$ and $\triangle DPQ$ respectively. Since

$$AF \times BF = QF \times DF,$$

the powers of $F$ with respect to $(ABC)$ and $(DPQ)$ are equal. Thus, $F$ lies on the radical axis of the two circles (the radical axis exists since the circles cannot be the same). Similarly, $E$ lies on the radical axis. Therefore, $OO' \perp EF$, which implies $OO' \perp BC$. Thus, $OO'$ is the perpendicular bisector of $BC$. It is then obvious that $O'X = O'D$, and hence $X$ lies on $(DPQ)$. $\square$

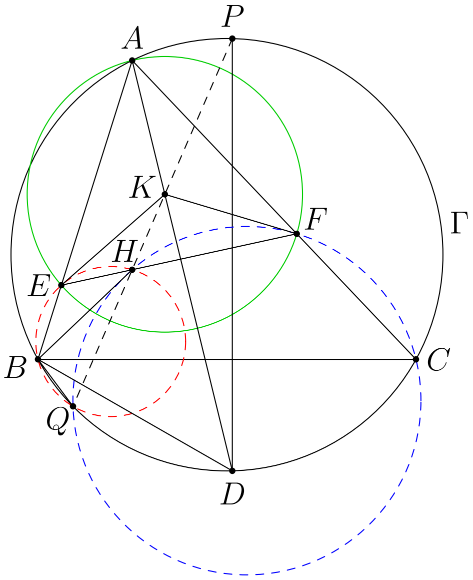
**Problem 7.** (USA TSTST 2019 Problem 5) Let $ABC$ be an acute triangle with orthocentre $H$ and circumcircle $\Gamma$. A line through $H$ intersects segments $AB$ and $AC$ at $E$ and $F$, respectively. Let $K$ be the circumcentre of $\triangle AEF$, and suppose line $AK$ intersects $\Gamma$ again at a point $D$. Prove that line $HK$ and the line through $D$ perpendicular to $BC$ meet on $\Gamma$.

**Solution.** Let $P$ be the point on $\Gamma$ such that $PD \perp BC$. Let $PH$ meet $\Gamma$ again at $Q$. We have to show that $K$ lies on $PHQ$. Firstly, since

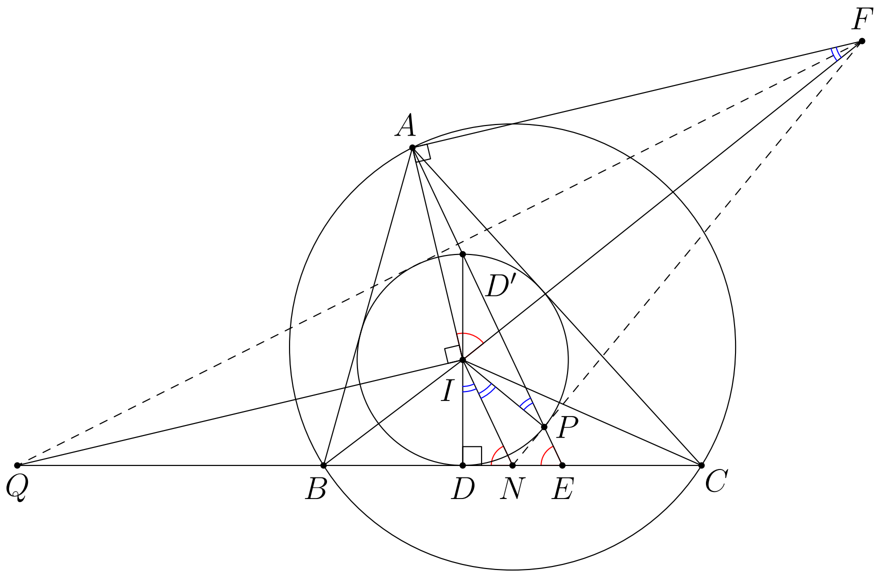
$$\angle BQH = \angle BDP = 90^\circ - \angle DBC = 90^\circ - \angle KAF = \angle FEA,$$

the points $B, E, H, Q$ are concyclic. By symmetry, $C, F, H, Q$ are concyclic. Therefore, $HQ$ is the radical axis of $(BEHQ)$ and $(CFHQ)$.

Next, since $\angle KEH = 90^\circ - \angle EAF = \angle HBA$, the line $EK$ is tangent to $(BEHQ)$. Similarly, $FK$ is tangent to $(CFHQ)$. As $KE = KF$, the powers of $K$ with respect to the two circles are equal. This shows $K$ lies on the radical axis $HQ$ of these circles as desired. $\square$




**Problem 8.** (Iran TST 2022) In triangle $ABC$, with $AB < AC$, $I$ is the incentre, $E$ is the intersection of $A$-excircle and $BC$. Point $F$ lies on the external angle bisector of $\angle BAC$ such that $E$ and $F$ lie on the same side of the line $AI$ and $\angle AIF = \angle AEB$. Point $Q$ lies on $BC$ such that $\angle AIQ = 90°$. Circle $\omega_b$ is tangent to $FQ$ and $AB$ at $B$, circle $\omega_c$ is tangent to $FQ$ and $AC$ at $C$ and both circles pass through the inside of triangle $ABC$. If $M$ is the midpoint of the arc $BC$ of the circumcircle of triangle $ABC$ which does not contain $A$, prove that $M$ lies on the radical axis of $\omega_b$ and $\omega_c$.




**Solution.** The first step is to prove that $FQ$ is tangent to the incircle. Let $D$ be the touching point of the incircle with side $BC$. Let $D'$ be the point such that $DD'$ is a diameter of the incircle. It is well-known that $A, D', E$ are collinear. Let $AD'$ meet the incircle again at $P$, and let $N$ be the midpoint of $BC$. Note that $N$ is also the midpoint of $DE$.

Firstly, since $\angle DPE = 90^\circ$ and $ND = NE$, we have $NP = ND = NE$. Therefore, $NP$ is tangent to the incircle and hence $\angle IPN = 90^\circ$. Secondly, by the midpoint theorem, $IN//D'E$. This implies

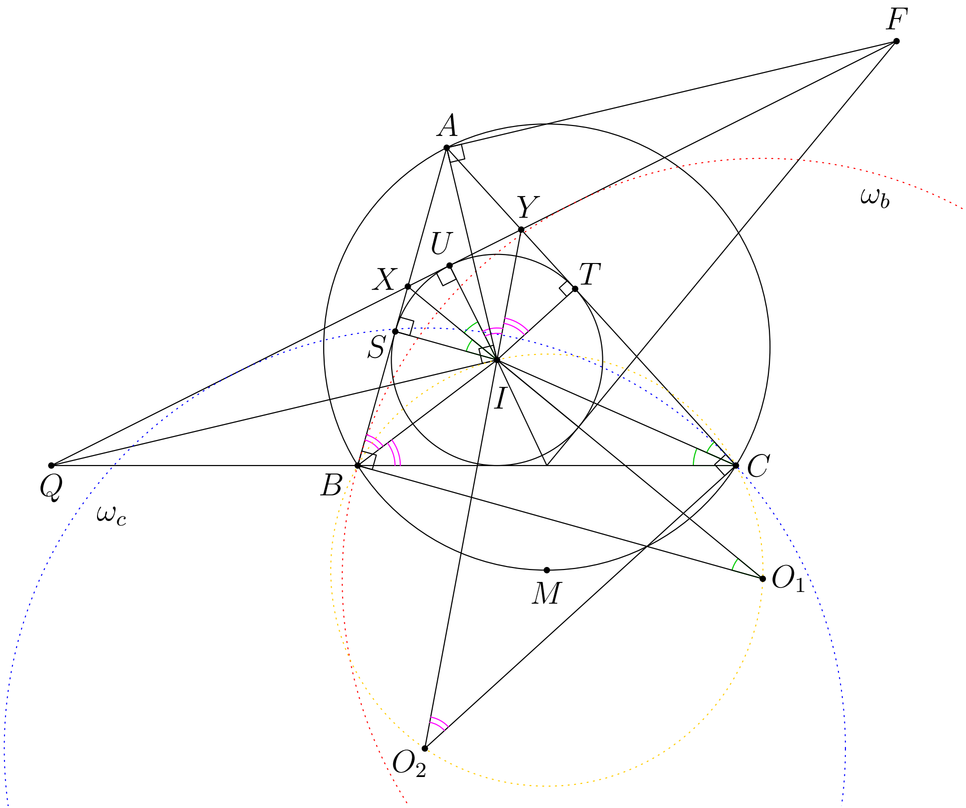
$$\angle API = \angle NIP = \angle DIN = 90^\circ - \angle IND = 90^\circ - \angle AEB = 90^\circ - \angle FIA = \angle AFI.$$

Therefore, $A, I, P, F$ are concyclic. This yields $\angle IPF = 180^\circ - \angle IAF = 90^\circ$, so $N, P, F$ are collinear. In other words, $NF$ is tangent to the incircle at $P$.

Next, as $I$ lies on the internal angle bisector of $\angle FNQ$ and

$$\angle FIQ = 90^\circ + \angle FIA = 90^\circ + \angle QNI,$$

we know that $I$ is the incentre of $\triangle FQN$. Therefore, $FQ$ is tangent to the incircle as claimed.




The second step is to show that $\omega_b$ and $\omega_c$ have the same radius. Let $O_1$ and $O_2$ be the centres of $\omega_b$ and $\omega_c$ respectively. Let $S, T, U$ be the contact points of the incircle with $AB, AC, FQ$ respectively. Suppose $AB$ and $AC$ meet $FQ$ at $X$ and $Y$ respectively. From

$$\begin{aligned}\angle TIU = \angle AYX = \angle CQY + \angle YCQ = 2\angle CQI + C = 2(180^\circ - \angle QIC - \angle ICQ) + C \\ = 360^\circ - 2\left(180^\circ - \frac{B}{2}\right) - 2 \cdot \frac{C}{2} + C = B,\end{aligned}$$

we obtain $\angle TIY = \angle UIY = \dfrac{B}{2}$. Similarly, $\angle SIX = \angle UIX = \dfrac{C}{2}$. So $\triangle SIB \sim \triangle TYI$ and $\triangle TIC \sim \triangle SXI$. This yields

$$\frac{XS}{SB} = \frac{XS}{SI} \times \frac{IS}{SB} = \frac{IT}{TC} \times \frac{YT}{TI} = \frac{YT}{TC}.$$

Note that $\omega_b$ can be obtained from the incircle via a homothety with centre $X$ and ratio $\dfrac{XB}{XS}$. Similarly, $\omega_c$ can be obtained from the incircle via a homothety with centre $Y$ and ratio $\dfrac{YC}{YT}$. From above, we have $\dfrac{XB}{XS} = \dfrac{YC}{YT}$. Therefore, $\omega_b$ and $\omega_c$ have the same size. It follows that their radical axis is the perpendicular bisector of $O_1O_2$.

Lastly, we prove that $O_1$ and $O_2$ lie on $(BCI)$. Since it is well-known that $M$ is the centre of $(BCI)$, this implies $M$ lies on the perpendicular bisector of $O_1O_2$ as desired.

Indeed, as $\angle O_1BA = \angle ISA = 90°$, we have $BO_1 // SI$ and hence

$$\angle IO_1B = \angle XIS = \angle ICB.$$

This implies $O_1$ lies on $(BCI)$. Similarly, $O_2$ also lies on $(BCI)$ and we are done. $\square$